Grating Lobe Suppression and Angle Estimation Based on Virtual Antennas Filling in Sparse Array

Jiangyou Zhu, Peng Lei, Jun Wang, Yaotian Zhang, and Changshun Yuan

***Abstract*—When the radar antenna spacing is larger than half wavelength, grating lobes appear in its antenna beam pattern. They could cause the ambiguity in the angle estimation of radar targets due to the similarity of amplitudes between grating lobes and the main lobe. To address such issue, this letter proposes a grating lobe suppression and angle estimation approach for sparse arrays by use of virtual antennas filling. It takes advantage of the distribution characteristics of grating lobes, and first obtains rough estimates of target angles. The least squares method is then employed to generate signals received by virtual antenna elements. Accordingly, the gaps between antenna elements of sparse array could be filled with those virtual ones to form an equivalent uniform full array. Hence, it achieves the grating lobe suppression as well as accurate estimation of target angles. Both simulations and experiments are carried out to verify its effectiveness in different types of sparse arrays.**

***Index Terms*—Angle estimation, grating lobe suppression, least squares estimation, sparse array.**

I. Introduction

THERE are two kinds of methods to suppress grating lobes in sparse array, one of which is to change the array layout, such as nested array and coprime array. The disadvantage of this method is that the effect of grating lobe suppression is not significant, which can only achieve modest grating lobe suppression. For example, Goffer et al. [1] used a sparse array with 128 antenna elements, but the grating lobe is only reduced by about 6 dB. Alshammary et al. [2] used the method of rotating irregular subarray to suppress grating lobes, but it increased the size of the subarray and had higher requirements for the irregular outer edge of the array, which is complicated and needs more antenna elements.

About another kind of methods, there are two main ways adopted by researchers to suppress grating lobes: beamforming and filling virtual antennas. As to beamforming, Feng et al. [3] proposed a method of randomly placing antenna elements at the transmitter, changing the size of antennas, and combining the digital beamforming and virtual antenna elements at the receiver. But it increased the complexity of hardware and was limited by the number of antenna elements. The second way is virtual antennas filling. For example, Feng et al. [4] filled virtual antenna elements in sparse array. Its disadvantage is that the method requires same number of antenna elements in each subarray and same spacing between two subarrays. The antenna spacing in the subarray is less than half wavelength, which is not applicable to sparse array with antenna spacing longer than half wavelength. To solve this problem, Chen et al. [5] used the method of minimum weighted norm and super spatially variant apodization to filling virtual antennas, but it involves large computation load and many snapshots. Besides, Wang et al. [6] filled virtual antanna elements in a specific sparse array without considering noise. Virtual antennas filling method proposed by Chang et al. [7] needs to know the target angle in advance, which is not suitable for angle estimation of unknown returned signals. For sparse arrays with longer antenna spacing, Kumar et al. [8] used a gap-filling method to reduce grating lobes; while the suppression effect was limited due to the fixed positions of antenna elements. Krivosheev et al. [9] and Yu et al. [10] designed complex array using massive antenna elements. On the contrary, our method is not only limited by the positions of antenna elements, but also not require massive antenna elements and complex array geometries.

This letter is organized as follows. In Section II, we analyze the forming causes and characteristics of grating lobes, and present the detail and process of proposed method. In Section III, the reliability of our method is verified by experimental results of both periodic and aperiodic antenna spacing. Finally, Section IV concludes this letter.

II. Proposed Method

A. Analysis of Causes and Characteristics of Grating Lobes

Let us consider the linear array, as shown in Fig. 1, where ξ_i is the position of the ith antenna element along the array axis. The far-field radiation function of the nth antenna element in the

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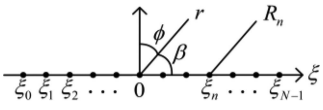


Fig. 1. Linear array composed of arbitrary antennas.

array can be expressed as

$$E_n = I_n e^{j\alpha_n} f_0(\theta,\varphi)\frac{e^{-jkR_n}}{R_n}, k = \frac{2\pi}{\lambda} \tag{1}$$

where I_n and α_n are the amplitude and the phase of the received signal of the nth antenna element, respectively, and $f_0(\theta,\varphi)$ is the pattern function of symmetric oscillator antenna element. λ is the wavelength of returned signal. The far radiation field of the array is determined as

$$\begin{aligned} E_T &= f_0(\theta,\varphi)\sum_{n=0}^{N-1} I_n e^{j\alpha_n}\frac{e^{-jkR_n}}{R_n} \\ &= \frac{e^{-jkr}}{r} f_0(\theta,\varphi)\sum_{n=0}^{N-1} I_n e^{j\alpha_n} e^{-jk(R_n-r)} \end{aligned} \tag{2}$$

where $(R_n - r)$ is the wave path difference, and N is the number of antenna elements. (2) can be reformulated as

$$E_T = \frac{e^{-jkr}}{r} f_0(\theta,\varphi)\cdot S(\beta) \tag{3}$$

where the array factor can be expressed as

$$S(\beta) = \sum_{n=0}^{N-1} \dot{I}_n e^{-j(k\xi_n\cos\beta+\alpha_n)}. \tag{4}$$

The phase changes uniformly. For uniform linear array, the array factor can be obtained as

$$S(u) = \sin\left(\frac{Nu}{2}\right), u = kd_r\cos\beta + \alpha \tag{5}$$

where d_r is the antenna spacing and $S(u)$ is a periodic function with period of 2π. When $u = 2m\pi, m = 0, \pm1, \pm2, \ldots$, the maximum value of (5) appears periodically.

When $m = 0$ and $u = 0$, the main lobe is directed at the angle of β_m, i.e., which is shown as

$$\beta_m = \arccos\left(-\frac{\alpha}{kd_r}\right). \tag{6}$$

When $m = \pm1, \pm2, \ldots$ and $u = kd_r(\cos\beta - \cos\beta_m) = \pm2\pi$, the corresponding grating lobe could be located.

Since $\beta \in [0,\pi]$ and $|\cos\beta - \cos\beta_m|_{\max} = 1 + |\cos\beta_m|$, the condition without grating lobes is given by

$$d_r < \frac{\lambda}{1+|\cos\beta_m|}. \tag{7}$$

1) Positions of Grating Lobes: When $m \neq 0$, the maximum value of (5) corresponds to the location of grating lobe and u can be expressed as

$$\begin{aligned} &u = kd_r(\cos\beta - \cos\beta_m) = 2m\pi, \\ &m = 0, \pm1, \pm2, \ldots, \beta \in [0,\pi]. \end{aligned} \tag{8}$$

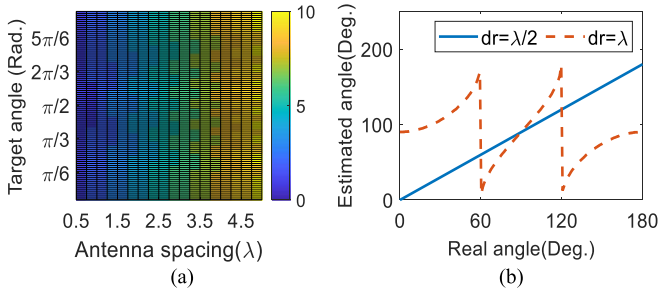


Fig. 2. (a) Number of grating lobes. (b) Relationship between real angle and estimated angle.

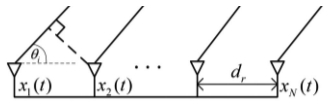


Fig. 3. Illustration of a linear sparse array.

Thus, the direction of grating lobe is

$$\beta = \arccos\left(m\frac{\lambda}{d_r} + \cos\beta_m\right). \tag{9}$$

2) Number of Grating Lobes: If $d_r = a\lambda$, $a > 0$, according to the characteristics of the inverse cosine function in (9), we can abtain

$$a(-1-\cos\beta_m) \le m \le a(1-\cos\beta_m). \tag{10}$$

When $m = 0$, β is the angle of the main lobe. When $m = \pm1, \pm2, \ldots$, β is the angle of the grating lobe.

Thus, the number of grating lobes for a given sparse array can be determined. The number of grating lobes with different antenna spacing and target signal is shown in Fig. 2(a).

B. Rough Estimation of Target Angles

Using (9), we obtain the positions of grating lobes. Taking all possible angles into consideration, the influence of grating lobes on the estimated results can be illustrated in Fig. 2(b), which is compared with the measured results to roughly estimate the angle of returned signal from targets.

C. Virtual Antennas Filling

The linear sparse array is shown in Fig. 3, and the model of returned signal is $\mathbf{X} = \mathbf{AS} + \mathbf{Y}$, which can be expressed as

$$\mathbf{X} = [\mathbf{X_1}\quad \mathbf{X_2}\quad \cdots\quad \mathbf{X_N}]^T \tag{11}$$

$$\mathbf{A} = \begin{bmatrix} e^{-j\omega_0\tau_{11}} & e^{-j\omega_0\tau_{12}} & \cdots & e^{-j\omega_0\tau_{1M}} \\ e^{-j\omega_0\tau_{21}} & e^{-j\omega_0\tau_{22}} & \cdots & e^{-j\omega_0\tau_{2M}} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-j\omega_0\tau_{N1}} & e^{-j\omega_0\tau_{N2}} & \cdots & e^{-j\omega_0\tau_{NM}} \end{bmatrix} \tag{12}$$

$$\mathbf{S} = [\mathbf{s_1}\quad \mathbf{s_2}\quad \cdots\quad \mathbf{s_M}]^T \tag{13}$$

$$\mathbf{n} = [\mathbf{n_1}\quad \mathbf{n_2}\quad \cdots\quad \mathbf{n_N}]^T \tag{14}$$

$$\omega_0\tau = 2\pi f\tau = 2\pi\frac{c}{\lambda}\tau = 2\pi\frac{d_r\cos\theta_i}{\lambda} \tag{15}$$

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where $\mathbf{X_i}$, $\mathbf{s_i}$, and $\mathbf{n_i}$ are column vectors of size $q \times 1$. $\mathbf{A}$ is the steering vector matrix. $\mathbf{S}$ is the returned signals. $\mathbf{n}$ is the Gaussian white noise. M is the number of targets. q is the number of snapshots. ω_0 is the angular frequency of the returned signal. τ_{mn} is the delay of the *m*th returned signal to the *n*th antenna element, which is relative to the reference antenna element. c is the speed of light. f is the frequency of returned signal. θ_i is the angle between the *i*th returned signal and the array axis.

Using the least squares method [7], we estimate the returned signals of the virtual antenna elements. The scheme of how to fill virtual antenna elements in sparse array is given in the following.

First, the conversion model of matrix from real antenna elements to virtual antenna elements is established as

$$\mathbf{A_v} = \mathbf{A_r C} + \mathbf{n_{LS}} \tag{16}$$

$$\mathbf{A_v} = \left[e^{-j2\pi \frac{d_v \cos\theta_1}{\lambda}} \; e^{-j2\pi \frac{d_v \cos\theta_2}{\lambda}} \; \cdots \; e^{-j2\pi \frac{d_v \cos\theta_q}{\lambda}} \right]^T \tag{17}$$

$$\mathbf{A_r} = \begin{bmatrix} e^{-j2\pi \frac{0 \cdot d_r \cos\theta_1}{\lambda}} & e^{-j2\pi \frac{1 \cdot d_r \cos\theta_1}{\lambda}} & \cdots & e^{-j2\pi \frac{(N-1) d_r \cos\theta_1}{\lambda}} \\ e^{-j2\pi \frac{0 \cdot d_r \cos\theta_2}{\lambda}} & e^{-j2\pi \frac{1 \cdot d_r \cos\theta_2}{\lambda}} & \cdots & e^{-j2\pi \frac{(N-1) d_r \cos\theta_2}{\lambda}} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-j2\pi \frac{0 \cdot d_r \cos\theta_q}{\lambda}} & e^{-j2\pi \frac{1 \cdot d_r \cos\theta_q}{\lambda}} & \cdots & e^{-j2\pi \frac{(N-1) d_r \cos\theta_q}{\lambda}} \end{bmatrix} \tag{18}$$

$$\mathbf{C} = [c_1 \quad c_2 \quad \cdots \quad c_N]^T \tag{19}$$

$$\mathbf{n_{LS}} = [n_1 \quad n_2 \quad \cdots \quad n_q]^T \tag{20}$$

$$\hat{\mathbf{C}} = (\mathbf{A_r}^T \mathbf{A_r})^{-1} \mathbf{A_r}^T \mathbf{A_v} \tag{21}$$

where $\mathbf{n_{LS}}$ is a column vector of size $q \times 1$. $\mathbf{A_r}$ is the steering vector of real antenna elements. $\mathbf{A_v}$ is the steering vector of virtual antenna elements. d_v is the virtual antenna spacing. $\mathbf{C}$ is the conversion matrix from real antenna elements to virtual antenna elements, and $\hat{\mathbf{C}}$ is the estimation of $\mathbf{C}$.

Second, the estimated returned signal is expressed as

$$\mathbf{X_{v_1}} = \mathbf{X}^T \hat{\mathbf{C}} \tag{22}$$

where $\mathbf{X_{v_1}}$ is the estimated signal of the first virtual antenna element and $\mathbf{X}$ is the returned signal matrix of the real antenna elements.

Third, on the basis, the estimated signal matrix of all virtual antenna elements can be abtained as

$$\mathbf{X_v} = [\mathbf{X_{v_1}} \quad \mathbf{X_{v_2}} \quad \cdots \quad \mathbf{X_{vv}}]^T \tag{23}$$

where V is the number of virtual antenna elements.

Finally, the uniform full array could be obtained after filling virtual antennas, and the signal of filled sparse array X_{all} is

$$\mathbf{X_{all}} = [\mathbf{X_1} \quad \mathbf{X_{v_1}} \quad \cdots \quad \mathbf{X_N} \quad \mathbf{X_{vv}}]^T. \tag{24}$$

III. Experimental Results

In experiments, we use the Texas Instruments millimeter-wave (mmW) radar sensor. It is composed of MMWCAS-DSP-EVM [11] and MMWCAS-RF-EVM [12], which includes four AWRx devices and the embedded antenna arrays. The spacing between two adjacent antenna elements is divided into two cases according to whether it is periodic or not. This section verifies that this method can be effectively applied to sparse arrays to solve angle ambiguity problem. Table I shows some parameter settings used in mmW radar-based experiments.

TABLE I
Parameters Settings in Experiments

Carrier frequency	Sampling rate	Modulation slope	q	N	Points of FFT
77 GHz	8 MHz	79 MHz/μs	256	4	256

TABLE II
Simulation Results with Different Target Angles

Serial number	Target angle(°)	Estimated mean(°)	Relative error
1	10	10.11	1.11%
2	20	20.14	0.69%
3	25	25.08	0.30%
4	30	30.05	0.17%
5	40	40.13	0.31%
6	50	50.18	0.36%
7	60	60.13	0.22%
8	70	70.11	0.16%
9	80	80.22	0.28%
10	90	90.22	0.25%
11	100	100.22	0.29%
12	110	110.35	0.32%
13	120	120.39	0.32%
14	130	130.47	0.36%
15	140	140.61	0.44%
16	150	150.85	0.57%
17	160	161.14	0.71%
18	170	172.67	1.57%

TABLE III
Comparison of Experimental Results Between the Uniform Full Array and the Sparse Array With Periodic Antenna Spacing

Real angles of targets	Results using uniform full array	Results using sparse array with virtual antennas filling	Relative errors of angle estimates
60°	60.13°	61.67°	2.78%
120°	120.40°	119.87°	0.11%

A. Periodic Antenna Spacing

The sparse array in this experiment has four antenna elements, and the spacing between two adjacent antenna elements is λ. Thus, it is needed to insert one virtual antenna element between two adjacent real ones.

Table II gives simulation results with a variety of target angles in the case. One hundred runs are conducted at each target angle. It not only includes the means of the estimated angles, but also the relative errors of estimates. The relative errors are less than 1.57% in the range of 10°–170°. Therefore, the feasibility and extraordinary stability of this method could be verified.

Furthermore, Figs. 4 and 5 are the experimental results with target angles of 60° and 120°, respectively, by using the mmW radar sensor. We use a corner reflector as the target, which is placed in different azimuthal directions. It can be seen from Table III that when the target angle is 60°, the difference between the result using sparse array with virtual antennas filling and the real angle is 1.67°, and the relative error is 2.78%. When the target angle is 120°, the difference between the result using sparse array with virtual antennas filling and the real angle is 0.13°, and the relative error is 0.11%.

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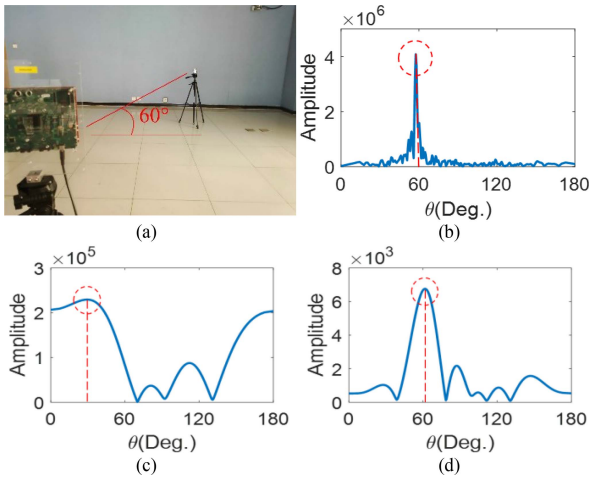


Fig. 4. Experimental results of angle estimation with periodic antenna spacing and the target angle of 60°. (a) Experimental scene. (b) Results using uniform full array. (c) Results using original sparse array. (d) Results using sparse array with virtual antennas filling.

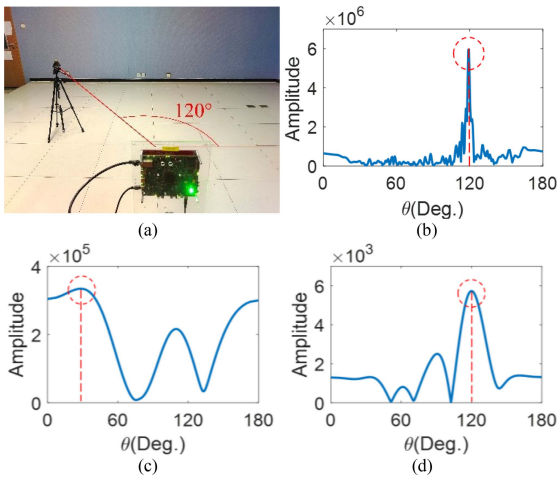


Fig. 5. Experimental results of angle estimation with periodic antenna spacing and the target angle of 120°. (a) Experimental scene. (b) Results using uniform full array. (c) Results using original sparse array. (d) Results using sparse array with virtual antennas filling.

B. Aperiodic Antenna Spacing

In this experiment, the total number of antenna elements is still four, and the spacing between two adjacent antenna elements is 1λ, 2λ, and 3λ, respectively. First, we establish an estimation model. Second, the least squares method is used to estimate the returned signals of the virtual antenna elements. Third, the sparse array with aperiodic antenna spacing could be filled to be an equivalent uniform linear array with antenna spacing of half wavelength. Thus, the number of virtual antenna elements that need to be inserted between two adjacent antenna elements is 1, 3, and 5, respectively. Finally, we use FFT to obtain the accurate estimation of returned signals. The experimental results using the sparse array with aperiodic antenna spacing in cases of different target angles are shown in Figs. 6 and 7.

Comparing Tables III and IV, the relative errors of angle estimates is increased due to the errors of least squares solutions for

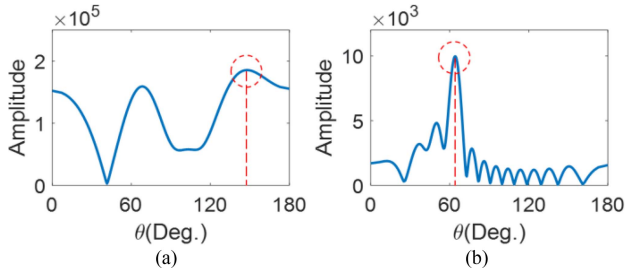


Fig. 6. Experimental results of angle estimation with aperiodic antenna spacing and the target angle of 60°. (a) Results using original sparse array. (b) Results using sparse array with virtual antennas filling.

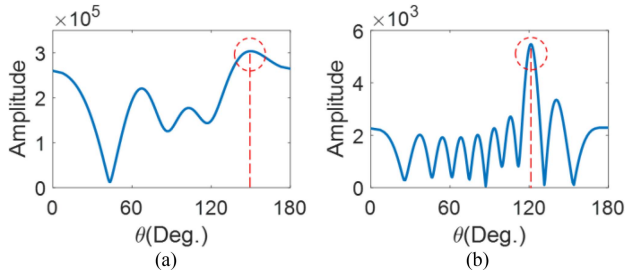


Fig. 7. Experimental results of angle estimation with aperiodic antenna spacing and the target angle of 120°. (a) Results using original sparse array. (b) Results using sparse array with virtual antennas filling.

TABLE IV
COMPARISON OF EXPERIMENTAL RESULTS BETWEEN THE UNIFORM FULL ARRAY AND THE SPARSE ARRAY WITH APERIODIC ANTENNA SPACING

Real angles of targets	Results using uniform full array	Results using sparse array with virtual antennas filling	Relative errors of angle estimates
60°	60.13°	64.20°	7.00%
120°	120.40°	124.10°	3.42%

virtual antenna elements in sparse array with aperiodic antenna spacing. However, the proposed method is still able to suppress grating lobes effectively in this case, and achieves the angle estimation with relative errors of 7.00% for the target angle of 60° and 3.42% for the target angle of 120°, respectively.

IV. CONCLUSION

In this letter, the characteristics of grating lobes in sparse radar arrays are explored, and a grating lobe suppression and angle estimation approach is proposed to handle the angle ambiguity problem caused by them. According to the angular relationship between the sparse array and its corresponding uniform full one, the estimates of target angles from original returned signals of real antennas could roughly be corrected. On the basis, these rough estimates and the least squares method are used to reckon signals from virtual antenna elements. By filling virtual antennas, we could obtain the signals received from an equivalent uniform full array, and then achieve the accurate estimation of target angles. Results on simulated and measured millimeter-wave radar data indicate that the proposed approach could mitigate the influence grating lobes in sparse arrays and improve the accuracy of angle estimation in both periodic and aperiodic sparse arrays.

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